

Freestanding Feature-Test Macros and Implementation-Defined Extensions

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Audience: Library Evolution Working Group

Change history

R0

- Branching from [P1641R3](#) and [P1642R3](#).
- Freestanding feature-test macros now required to be present in hosted.
- Added examples and use cases.
- Elaborating on how an implementation can extend the bare minimum of freestanding.

Introduction

Users of freestanding implementations would like to be able to detect which library facilities are available for use. With the C++20 feature-test macros, all the feature-test macros are required to be defined in both freestanding and hosted implementations, even when the associated hosted facilities are not available. This paper will make the presence of feature-test macros for hosted facilities implementation-defined. This paper will also bump the version number on all library feature-test macros, so that extensions can be detected.

This paper also clarifies that freestanding implementations need not stop at the bare minimum support for freestanding. Freestanding implementations may provide additional facilities from the standard library, so long as those facilities meet the associated hosted requirements.

Motivation and Design

Beyond the bare minimum

This paper grants implementations the freedom to include more than just the facilities that are marked `//freestanding`. There will be platforms where, for example, floating point is both available and frequently used (e.g. GPU environments). On those platforms, it is desirable for implementations to provide many of the facilities in the `<cmath>` and `<complex>` headers. We must be careful here though, as we don't want to allow divergence of implementation, and we want to be able to add facilities to the required subset of freestanding. To permit both vendor extension and future freestanding growth, this paper will require that additionally included facilities must meet the same requirements as for a hosted implementation, with a few exceptions.

Freestanding implementations may include additional macros and namespace scoped entities, beyond what is required in a minimal freestanding implementation. A freestanding implementation does not need to provide the entirety of a header. If a freestanding implementation supplies an additional class, the entirety of the class must be present and meet all the hosted requirements. This prevents some useful vendor extensions, notably `at()`-less array, `value()`-less optional, and similarly partial `string_view` and `bitset`. This does not prevent us from standardizing a partial class in the future, though it does prevent us from getting usage experience in a conforming implementation.

Freestanding implementations may make the use of any non-freestanding, namespace scoped function or function template ill-formed (e.g. generally by marking it `=delete`). The intent is to allow the freestanding implementation to control overload resolution, so that a library that works with a freestanding implementation will have the same semantics in a hosted implementation.

Users will be able to rely on the freestanding portions of the standard library. The freestanding portions have portable semantics, and are required to be present. Users may then decide to rely on implementation-defined additions. WG21 will be able to add hosted entities to the freestanding subset without little fear of breaking implementation-defined additions, because the implementation-defined additions needed to meet the hosted requirements.

Library feature-test macro bump

C++20 requires that all implementations (freestanding and hosted) define all of the library feature-test macros in the `<version>` header, even the feature-test macros that correspond to facilities not required to be present in freestanding implementations. This means that those feature-test macros provide misleading results on freestanding implementations. This isn't just a theoretical problem. [Existing implementations](#) are already deploying `<version>` headers that report support for `std::filesystem` facilities, `std::chrono` facilities, and many others, even though those feature-test macros indicate support for features that require the support of an operating system.

In order for users of freestanding implementations to be able to detect extensions of freestanding, the users need a way of distinguishing the C++20 macro requirements from an accurate expression of extension. This paper will bump all library macro versions, so that users can distinguish between these cases.

Freestanding feature-test macros

The following macros are now required to be present in freestanding implementations. The other library feature test macros in the `<version>` header are not required to be present on freestanding implementations. The corresponding facilities were either required to be present in C++20, or are added in P1642.

- `__cpp_lib_addressof_constexpr`
- `__cpp_lib_allocator_traits_is_always_equal`
- `__cpp_lib_apply`
- `__cpp_lib_as_const`
- `__cpp_lib_assume_aligned`
- `__cpp_lib_atomic_flag_test`
- `__cpp_lib_atomic_float`
- `__cpp_lib_atomic_is_always_lock_free`
- `__cpp_lib_atomic_ref`
- `__cpp_lib_atomic_value_initialization`
- `__cpp_lib_atomic_wait`
- `__cpp_lib_bind_front`
- `__cpp_lib_bit_cast`
- `__cpp_lib_bitops`
- `__cpp_lib_bool_constant`
- `__cpp_lib_bounded_array_traits`
- `__cpp_lib_byte`
- `__cpp_lib_char8_t`
- `__cpp_lib_concepts`
- `__cpp_lib_constexpr_functional`
- `__cpp_lib_constexpr_iterator`
- `__cpp_lib_constexpr_memory`
- `__cpp_lib_constexpr_tuple`
- `__cpp_lib_constexpr_utility`
- `__cpp_lib_destroying_delete`
- `__cpp_lib_endian`
- `__cpp_lib_exchange_function`
- `__cpp_lib_hardware_interference_size`
- `__cpp_lib_has_unique_object_representations`
- `__cpp_lib_int_pow2`
- `__cpp_lib_integer_sequence`
- `__cpp_lib_integral_constant_callable`
- `__cpp_lib_invoke`
- `__cpp_lib_is_aggregate`
- `__cpp_lib_is_constant_evaluated`
- `__cpp_lib_is_final`
- `__cpp_lib_is_invocable`
- `__cpp_lib_is_layout_compatible`
- `__cpp_lib_is_nothrow_convertible`
- `__cpp_lib_is_null_pointer`
- `__cpp_lib_is_pointer_interconvertible`
- `__cpp_lib_is_swappable`
- `__cpp_lib_laundry`
- `__cpp_lib_logical_traits`
- `__cpp_lib_make_from_tuple`
- `__cpp_lib_make_reverse_iterator`
- `__cpp_lib_nonmember_container_access`
- `__cpp_lib_not_fn`
- `__cpp_lib_null_iterators`
- `__cpp_lib_remove_cvref`
- `__cpp_lib_result_of_sfinae`
- `__cpp_lib_source_location`
- `__cpp_lib_ssize`
- `__cpp_lib_three_way_comparison`
- `__cpp_lib_to_address`
- `__cpp_lib_transformation_trait_aliases`
- `__cpp_lib_transparent_operators`
- `__cpp_lib_tuple_element_t`
- `__cpp_lib_tuples_by_type`
- `__cpp_lib_type_identity`
- `__cpp_lib_type_trait_variable_templates`
- `__cpp_lib_uncaught_exceptions`
- `__cpp_lib_unwrap_ref`
- `__cpp_lib_void_t`

New feature-test macros

Users of freestanding implementations will want to know whether including a formerly hosted-only header will work or not. Users of freestanding implementations will also want to know if all the facilities that are required to be in freestanding have been made available yet. This is a concern for

highly portable libraries, and for users that need to support old and new compilers.

These feature-test macros are provided at a per-header granularity. This enables implementations to advertise new capabilities more easily than a single feature-test macro for the entirety of this paper. This also follows the precedent set by the `__cpp_lib_constexpr_*` macros.

If new, pre-C++20 functionality is added to the freestanding subset of C++, then the respective feature-test macro for the header should be bumped. If the functionality is new in C++23 or later, then alternative approaches should be taken. These alternative approaches are discussed in the examples section.

Name	Header
<code>__cpp_lib_freestanding_utility</code>	<code><utility></code>
<code>__cpp_lib_freestanding_tuple</code>	<code><tuple></code>
<code>__cpp_lib_freestanding_ratio</code>	<code><ratio></code>
<code>__cpp_lib_freestanding_memory</code>	<code><memory></code>
<code>__cpp_lib_freestanding_functional</code>	<code><functional></code>
<code>__cpp_lib_freestanding_iterator</code>	<code><iterator></code>
<code>__cpp_lib_freestanding_ranges</code>	<code><ranges></code>

Partial macro coverage

The following, existing feature-test macros cover some features that I am making freestanding, and some features that I am not requiring to be freestanding. These feature-test macros won't be required in freestanding, as they could cause substantial confusion when the hosted parts of those features aren't available. The per-header `__cpp_lib_freestanding_*` macros should provide a suitable replacement in freestanding environments.

- `__cpp_lib_boyer_moore_searcher`: `default_searcher` is in, other searchers require the heap.
- `__cpp_lib_constexpr_dynamic_alloc`: `constexprification` of various memory algorithms is in, anything dealing with `std::allocator` is out
- `__cpp_lib_ranges`: stream iterators are out
- `__cpp_lib_raw_memory_algorithms`: `ExecutionPolicy` overloads are out

Feature-test macro policy recommendations

This paper patches up many of the historical problems with freestanding and feature-test macros. The following are the guidelines I recommend to keep feature-test macros useful for freestanding in the future.

- If a paper contains some facilities that are freestanding, and some that are hosted, then the paper needs at least two feature test macros. `__cpp_lib_foo` will cover the full paper in the hosted case, and `__cpp_lib_freestanding_foo` will cover the freestanding portions.
- If a paper adds facilities to the hosted implementation and has a feature-test macro, and a later paper adds the entirety of the paper to freestanding, then bump the version of the original feature-test macro.
- If a paper adds facilities to the hosted implementation and has a feature-test macro, and a later paper adds a portion of the paper to freestanding, then add a new freestanding macro for the feature, of the form `__cpp_lib_freestanding_foo`.

Examples

The following feature detection examples should ...

- ... work on freestanding, hosted, and with implementation-defined freestanding extensions.
- ... work with old standards and new standards.
- ... provide a conservative "false" answer when working with an old standard.
- ... not cause warnings for the use of an undefined macro, like with `-Wundef`.

Detect C++20 (or older) feature that is not required in freestanding

Is `filesystem::copy` available?

```
#if __STDC_HOSTED__ && defined(__cpp_lib_filesystem) && __cpp_lib_filesystem >= 201703L
// hosted success path
#elif defined(__cpp_lib_filesystem) && __cpp_lib_filesystem >= 202007L
// freestanding extension path
#else
// fallback path
#endif
```

Detect C++20 (or older) feature that is now entirely required in freestanding

Is `ssize` available?

```
#if __STDC_HOSTED__ && defined(__cpp_lib_ssize) && __cpp_lib_ssize >= 201902L
// hosted success path
#elif defined(__cpp_lib_ssize) && __cpp_lib_ssize >= 202007L
// freestanding path
#else
// fallback path
#endif
```

Detect C++20 (or older) feature that is now partially required in freestanding

Is the non-parallel version of uninitialized_default_construct available?

```
#if defined(__cpp_lib_raw_memory_algorithms) && __cpp_lib_raw_memory_algorithms >= 201606L
// hosted success path and freestanding extension path
#elif defined(__cpp_lib_freestanding_memory) && __cpp_lib_freestanding_memory >= 202007L
// freestanding path
#else
// fallback path
#endif
```

Detect pre-feature-test-macro feature that is required in freestanding

Is tuple available?

```
#if defined(__cpp_lib_freestanding_tuple) && __cpp_lib_freestanding_tuple >= 202007L
// freestanding and future hosted success path
#elif __STDC_HOSTED__
// flakey hosted success path. Assume tuple works here
#else
// fallback path
#endif
```

Detect pre-feature-test-macro feature that is not required in freestanding

Is thread available?

```
// No good answers here. Currently calling this out of scope.
```

Detect C++23 feature that is entirely freestanding in the initial paper

```
#if defined(__cpp_lib_always_freestanding_feature) && __cpp_lib_always_freestanding_feature >= 202202L
// hosted and freestanding success path
#else
// fallback path
#endif
```

Detect C++23 feature that is hosted in the initial paper, but made entirely freestanding later (possibly in another version of C++)

For this category of features, the recommendation will be to bump the integer constant in the version test macro.

```
#if defined(__cpp_lib_eventually_freestanding_feature)
  #if __cpp_lib_eventually_freestanding_feature >= 202702L
    // freestanding success path. Will also trigger for new hosted toolchains
  #elif __STDC_HOSTED__ && __cpp_lib_eventually_freestanding_feature >= 202102L
    // Interim hosted success path
  #else
    // Interim freestanding fallback path
  #endif
#else
// fallback path
#endif

// Alternative that will work 99% of the time
#if defined(__cpp_lib_eventually_freestanding_feature)
// freestanding and hosted success, probably
#else
// fallback path
#endif
```

Detect freestanding part of C++23 feature that is partially freestanding in the initial paper

For this category of features, the recommendation will be to have two macros, one for the freestanding portion, and one for the entire paper. The macros should follow the convention `__cpp_lib_foo` for the full paper, and `__cpp_lib_freestanding_foo` for the freestanding portion.

Is the freestanding portion of feature foo available?

```
#if defined(__cpp_lib_freestanding_foo) && __cpp_lib_freestanding_foo >= 202202L
// hosted and freestanding success path
#else
// fallback path
#endif
```

Detect freestanding part of C++23 feature that is made freestanding after the initial paper (possibly in a different C++ release)

For this category of features, the recommendation will be to add a new freestanding macro for the feature, following the convention of `__cpp_lib_freestanding_bar` for the freestanding portion.

Is the freestanding portion of feature bar available?

```
#if defined(__cpp_lib_bar) && __cpp_lib_bar >= 202202L
// Old hosted toolchain path and freestanding extension path
#elif defined(__cpp_lib_freestanding_bar) && __cpp_lib_freestanding_bar >= 202702L
// freestanding success path. Will also trigger for new hosted toolchains
#else
// fallback path
#endif
```

Rejected alternatives

Allow partial classes in conforming vendor extensions to freestanding

Allowing partial classes gives implementers a great deal of freedom to experiment and to provide extensions. It also permits abuse by the implementers. On the positive side, an implementation could provide `std::array` without `at`. This helps replace C facilities with better facilities from C++. On the negative side, an implementation could `=delete` the copy constructor. Removing the wrong functions would also mean that a class may not satisfy the same concepts in hosted vs. freestanding.

Update old feature-test macros to indicate freestanding status

Rather than have a `__cpp_lib_freestanding_ratio` (for example), we could instead update `__cpp_lib_chrono`. Users could test the value of `__cpp_lib_chrono` to determine if it is freestanding safe or not.

There are at least two big flaws with this approach.

First, not all facilities (particularly older ones) have existing feature-test macros to bump (e.g. `unique_ptr`). Users would still like to be able to detect the availability of the features in freestanding.

Second, implementers don't always implement features in the same order that they are added to the working draft. If a new feature were added to the `<chrono>` header that necessitated a feature-test macro bump, and an implementer addressed that feature before making the `<ratio>` header freestanding safe, then the implementer would either need to stick with the pre-freestanding feature-test macro version, or provide a misleading feature-test macro version.

Per-freestanding paper feature-test macro granularity

Rather than introduce a feature-test macro per header in this paper, I could instead introduce one feature-test macro... perhaps `__cpp_lib_freestanding`. Each paper that adds old facilities to freestanding could then introduce its own macro, or bump the old one.

This approach can work, but it restricts the order in which implementers can meaningfully implement freestanding features. All of the old paper needs to be done before any of the old paper's progress can be advertised. All of the old paper needs to be done before advertising any newer papers.

Per-facility feature-test macro granularity

This paper could choose to add a feature-test macro for `__cpp_lib_freestanding_unique_ptr`, `__cpp_lib_freestanding_pair`, and `__cpp_lib_freestanding_iterator_categories`. There would be some value to users in that they could express exactly what it is they need, and see if that very specific facility is available.

However, this approach is an implementation hassle, and prone to endless wg21 debates on how to partition and name the facilities. Grouping facilities by header provides a natural partitioning and naming scheme.

Rely on `__has_include`

Some have suggested using `__has_include` to detect whether the `<tuple>` header (for example) is usable on a particular freestanding implementation. This doesn't work in practice for multiple reasons.

The first example is the `<iterator>` header. In older versions of Visual Studio, a user could attempt to `#include <iterator>`. The streaming iterators in the header result in compiler errors. We need feature-test macros to indicate whether including the header is well-formed.

The second example is headers that do standard versions checks inside. The `libstdc++` implementation of the `<ranges>` header is mostly empty if the language version is less than C++20. `__has_include` will still report the header as present though.

Don't bump all the library feature-test macros

This makes it even more difficult to detect extensions to freestanding. [Existing implementations](#) report support for `__cpp_lib_filesystem` when the `-ffreestanding` flag is passed, even though it is unlikely that filesystem is supported on such platforms. The C++20 standard currently requires this behavior.

Make `__cpp_lib_freestanding_*` macros freestanding only

This complicates the client feature-test code for features that were marked freestanding in their initial papers. It doesn't make any of the client feature-test code any simpler. It also violates the principle that freestanding should be a subset of hosted.

Define `__cpp_lib_freestanding_*` macros to 0 in hosted

Users of the freestanding macros would still need to account for the case where the macro isn't present. Defining the macro to zero doesn't provide any information beyond what `__STDC_HOSTED__` provides.

Wording

Additions to [compliance]

Please append the following paragraphs to [compliance].

- 4 The *hosted library facilities* are the set of facilities described in this document that are required for hosted implementations, but not required for freestanding implementations. A freestanding implementation provides a (possibly empty) implementation-defined subset of the hosted library facilities. Unless otherwise specified, the requirements on each macro and namespace scoped entity provided in this way shall be the same as the corresponding requirements in a hosted implementation.
- 5 A freestanding implementation provides *deleted definitions* for a (possibly empty) implementation-defined subset of the namespace scoped functions and function templates from the hosted library facilities.
[*Note:* An implementation may provide a deleted definition so that overload resolution does not silently change when migrating a library from a freestanding implementation to a hosted implementation. -end note]

Change in [version.syn]

Please make the following changes to [version.syn]. Use the current year-based constant instead of "new-val".

```
#define __cpp_lib_addressof_constexpr 201603Lnew-val // freestanding, also in <memory>
#define __cpp_lib_allocator_traits_is_always_equal 201411Lnew-val
// freestanding, also in <memory>, <scoped-allocator>, <string>, <deque>, <forward-list>, <list>, <vector>,
// <map>, <set>, <unordered-map>, <unordered-set>
#define __cpp_lib_any 201606Lnew-val // also in <any>
#define __cpp_lib_apply 201603Lnew-val // freestanding, also in <tuple>
#define __cpp_lib_array_constexpr 201811Lnew-val // also in <iterator>, <array>
#define __cpp_lib_as_const 201510Lnew-val // freestanding, also in <utility>
#define __cpp_lib_assume_aligned 201811Lnew-val // freestanding, also in <memory>
#define __cpp_lib_atomic_flag_test 201907Lnew-val // freestanding, also in <atomic>
#define __cpp_lib_atomic_float 201711Lnew-val // freestanding, also in <atomic>
#define __cpp_lib_atomic_is_always_lock_free 201603Lnew-val // freestanding, also in <atomic>
#define __cpp_lib_atomic_lock_free_type_aliases 201907Lnew-val // also in <atomic>
#define __cpp_lib_atomic_ref 201806Lnew-val // freestanding, also in <atomic>
#define __cpp_lib_atomic_shared_ptr 201711Lnew-val // also in <memory>
#define __cpp_lib_atomic_value_initialization 201911Lnew-val // freestanding, also in <atomic>, <memory>
#define __cpp_lib_atomic_wait 201907Lnew-val // freestanding, also in <atomic>
#define __cpp_lib_barrier 201907Lnew-val // also in <barrier>
#define __cpp_lib_bind_front 201907Lnew-val // freestanding, also in <functional>
#define __cpp_lib_bit_cast 201806Lnew-val // freestanding, also in <bit>
#define __cpp_lib_bitops 201907Lnew-val // freestanding, also in <bit>
#define __cpp_lib_bool_constant 201505Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_bounded_array_traits 201902Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_boyer_moore_searcher 201603Lnew-val // also in <functional>
#define __cpp_lib_byte 201603Lnew-val // freestanding, also in <cstdint>
#define __cpp_lib_char8_t 201907Lnew-val
// freestanding, also in <atomic>, <filesystem>, <istream>, <limits>, <locale>, <ostream>, <string>, <string-view>
#define __cpp_lib_chrono 201907Lnew-val // also in <chrono>
#define __cpp_lib_chrono_udls 201304Lnew-val // also in <chrono>
#define __cpp_lib_clamp 201603Lnew-val // also in <algorithm>
#define __cpp_lib_complex_udls 201309Lnew-val // also in <complex>
#define __cpp_lib_concepts 202002Lnew-val // freestanding, also in <concepts>
#define __cpp_lib_constexpr_algorithms 201806Lnew-val // also in <algorithm>
#define __cpp_lib_constexpr_complex 201711Lnew-val // also in <complex>
#define __cpp_lib_constexpr_dynamic_alloc 201907Lnew-val // also in <memory>
#define __cpp_lib_constexpr_functional 201907Lnew-val // freestanding, also in <functional>
#define __cpp_lib_constexpr_iterator 201811Lnew-val // freestanding, also in <iterator>
#define __cpp_lib_constexpr_memory 201811Lnew-val // freestanding, also in <memory>
#define __cpp_lib_constexpr_numeric 201911Lnew-val // also in <numeric>
#define __cpp_lib_constexpr_string 201907Lnew-val // also in <string>
#define __cpp_lib_constexpr_string_view 201811Lnew-val // also in <string-view>
#define __cpp_lib_constexpr_tuple 201811Lnew-val // freestanding, also in <tuple>
#define __cpp_lib_constexpr_utility 201811Lnew-val // freestanding, also in <utility>
#define __cpp_lib_constexpr_vector 201907Lnew-val // also in <vector>
```

```

#define __cpp_lib_coroutine 201902Lnew-val // also in <coroutine>
#define __cpp_lib_destroying_delete 201806Lnew-val // freestanding, also in <new>
#define __cpp_lib_enable_shared_from_this 201603Lnew-val // also in <memory>
#define __cpp_lib_endian 201907Lnew-val // freestanding, also in <bit>
#define __cpp_lib_erase_if 202002Lnew-val
    // also in <string>, <deque>, <forward-list>, <list>, <vector>, <map>, <set>, <unordered-map>,
    // <unordered-set>
#define __cpp_lib_exchange_function 201304Lnew-val // freestanding, also in <utility>
#define __cpp_lib_execution 201902Lnew-val // also in <execution>
#define __cpp_lib_filesystem 201703Lnew-val // also in <filesystem>
#define __cpp_lib_format 201907Lnew-val // also in <format>
#define __cpp_lib_freestanding_functional new-val // freestanding, also in <functional>
#define __cpp_lib_freestanding_iterator new-val // freestanding, also in <iterator>
#define __cpp_lib_freestanding_memory new-val // freestanding, also in <memory>
#define __cpp_lib_freestanding_ranges new-val // freestanding, also in <ranges>
#define __cpp_lib_freestanding_ratio new-val // freestanding, also in <ratio>
#define __cpp_lib_freestanding_tuple new-val // freestanding, also in <tuple>
#define __cpp_lib_freestanding_utility new-val // freestanding, also in <utility>
#define __cpp_lib_gcd_lcm 201606Lnew-val // also in <numeric>
#define __cpp_lib_generic_associative_lookup 201304Lnew-val // also in <map>, <set>
#define __cpp_lib_generic_unordered_lookup 201811Lnew-val
    // also in <unordered-map>, <unordered-set>
#define __cpp_lib_hardware_interference_size 201703Lnew-val // freestanding, also in <new>
#define __cpp_lib_has_unique_object_representations 201606Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_hypot 201603Lnew-val // also in <cmath>
#define __cpp_lib_incomplete_container_elements 201505Lnew-val
    // also in <forward-list>, <list>, <vector>
#define __cpp_lib_int_pow2 202002Lnew-val // freestanding, also in <bit>
#define __cpp_lib_integer_comparison_functions 202002Lnew-val // also in <utility>
#define __cpp_lib_integer_sequence 201304Lnew-val // freestanding, also in <utility>
#define __cpp_lib_integral_constant_callable 201304Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_interpolate 201902Lnew-val // also in <cmath>, <numeric>
#define __cpp_lib_invoke 201411Lnew-val // freestanding, also in <functional>
#define __cpp_lib_is_aggregate 201703Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_is_constant_evaluated 201811Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_is_final 201402Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_is_invocable 201703Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_is_layout_compatible 201907Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_is_nothrow_convertible 201806Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_is_null_pointer 201309Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_is_pointer_interconvertible 201907Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_is_swappable 201603Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_jthread 201911Lnew-val // also in <stop-token>, <thread>
#define __cpp_lib_latch 201907Lnew-val // also in <latch>
#define __cpp_lib_laundry 201606Lnew-val // freestanding, also in <new>
#define __cpp_lib_list_remove_return_type 201806Lnew-val // also in <forward-list>, <list>
#define __cpp_lib_logical_traits 201510Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_make_from_tuple 201606Lnew-val // freestanding, also in <tuple>
#define __cpp_lib_make_reverse_iterator 201402Lnew-val // freestanding, also in <iterator>
#define __cpp_lib_make_unique 201304Lnew-val // also in <memory>
#define __cpp_lib_map_try_emplace 201411Lnew-val // also in <map>
#define __cpp_lib_math_constants 201907Lnew-val // also in <numbers>
#define __cpp_lib_math_special_functions 201603Lnew-val // also in <cmath>
#define __cpp_lib_memory_resource 201603Lnew-val // also in <memory-resource>
#define __cpp_lib_node_extract 201606Lnew-val
    // also in <map>, <set>, <unordered-map>, <unordered-set>
#define __cpp_lib_nonmember_container_access 201411Lnew-val
    // freestanding, also in <array>, <deque>, <forward-list>, <iterator>, <list>, <map>, <regex>, <set>, <string>,
    // <unordered-map>, <unordered-set>, <vector>
#define __cpp_lib_not_fn 201603Lnew-val // freestanding, also in <functional>
#define __cpp_lib_null_iterators 201304Lnew-val // freestanding, also in <iterator>
#define __cpp_lib_optional 201606Lnew-val // also in <optional>
#define __cpp_lib_parallel_algorithm 201603Lnew-val // also in <algorithm>, <numeric>
#define __cpp_lib_polymorphic_allocator 201902Lnew-val // also in <memory>
#define __cpp_lib_quoted_string_io 201304Lnew-val // also in <iomanip>
#define __cpp_lib_ranges 201911Lnew-val
    // also in <algorithm>, <functional>, <iterator>, <memory>, <ranges>
#define __cpp_lib_raw_memory_algorithms 201606Lnew-val // also in <memory>
#define __cpp_lib_remove_cvref 201711Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_result_of_sfinae 201210Lnew-val // freestanding, also in <functional>, <type-traits>
#define __cpp_lib_robust_nonmodifying_seq_ops 201304Lnew-val // also in <algorithm>
#define __cpp_lib_sample 201603Lnew-val // also in <algorithm>
#define __cpp_lib_scoped_lock 201703Lnew-val // also in <mutex>
#define __cpp_lib_semaphore 201907Lnew-val // also in <semaphore>
#define __cpp_lib_shared_mutex 201505Lnew-val // also in <shared-mutex>
#define __cpp_lib_shared_ptr_arrays 201707Lnew-val // also in <memory>
#define __cpp_lib_shared_ptr_weak_type 201606Lnew-val // also in <memory>
#define __cpp_lib_shared_timed_mutex 201402Lnew-val // also in <shared-mutex>
#define __cpp_lib_shift 201806Lnew-val // also in <algorithm>
#define __cpp_lib_smart_ptr_for_overwrite 202002Lnew-val // also in <memory>

```



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#define __cpp_lib_source_location 201907Lnew-val // freestanding, also in <source-location>
#define __cpp_lib_span 202002Lnew-val // also in <span>
#define __cpp_lib_ssize 201902Lnew-val // freestanding, also in <iterator>
#define __cpp_lib_starts_ends_with 201711Lnew-val // also in <string>, <string-view>
#define __cpp_lib_string_udls 201304Lnew-val // also in <string>
#define __cpp_lib_string_view 201803Lnew-val // also in <string>, <string-view>
#define __cpp_lib_syncbuf 201803Lnew-val // also in <syncstream>
#define __cpp_lib_three_way_comparison 201907Lnew-val // freestanding, also in <compare>
#define __cpp_lib_to_address 201711Lnew-val // freestanding, also in <memory>
#define __cpp_lib_to_array 201907Lnew-val // also in <array>
#define __cpp_lib_to_chars 201611Lnew-val // also in <charconv>
#define __cpp_lib_transformation_trait_aliases 201304Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_transparent_operators 201510Lnew-val // freestanding, also in <memory>, <functional>
#define __cpp_lib_tuple_element_t 201402Lnew-val // freestanding, also in <tuple>
#define __cpp_lib_tuples_by_type 201304Lnew-val // freestanding, also in <utility>, <tuple>
#define __cpp_lib_type_identity 201806Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_type_trait_variable_templates 201510Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_uncaught_exceptions 201411Lnew-val // freestanding, also in <exception>
#define __cpp_lib_unordered_map_try_emplace 201411Lnew-val // also in <unordered-map>
#define __cpp_lib_unwrap_ref 201811Lnew-val // freestanding, also in <type-traits>
#define __cpp_lib_variant 201606Lnew-val // also in <variant>
#define __cpp_lib_void_t 201411Lnew-val // freestanding, also in <type-traits>

```